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The use of EXOGAM for in-beam spectroscopy of proton drip-line nuclei with radioactive ion beams

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ABSTRACT

One of the first fusion–evaporation experiments using radioactive ion beams was performed at GANIL in order to study proton-rich nuclei of the light rare-earth region. The low production cross-section of the exotic species of interest, in combination with the low intensity of the beam and its induced background, demanded the use of a highly efficient experimental setup. This consisted of the EXOGAM γ-ray spectrometer coupled for the first time with both the DIAMANT charged-particle array and the VAMOS recoil spectrometer. In this report the experimental challenges of such studies will be discussed and the experimental achievements of the in-beam spectroscopy of proton drip-line nuclei using EXOGAM will be presented.

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1. Introduction

Proton drip-line nuclei provide an important basis for the understanding of fundamental nuclear interactions and the determination of the limits of nuclear stability. In particular, proton-rich rare-earth nuclei of the $A\sim130$ region are of great interest since highly deformed prolate ground states are expected. Indeed these nuclei, with neutron numbers lying midway between the $N=50$ and 82 shell closures and proton numbers far away from the $Z=50$ shell closure, are predicted to show maximal ground-state deformations of $\beta_2\sim0.4$. This value of quadrupole deformation corresponds to a prolate nuclear shape with a 3:2 axis ratio and is comparable to that deduced for superdeformed bands observed in cerium isotopes at high spin, allowing some very interesting comparisons to be made.

Mapping and understanding major regions of deformation away from closed-shells is an important aspect of nuclear structure physics. Experiments have succeeded in studying the nuclei approaching, but not yet reaching, the peak of this deformation in the light rare-earth region ($A\sim130$). The use of radioactive ion beams (RIBs) opens up this exotic region and allows the probing of nuclei at and around the peak of this systematic feature.

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In the present report results from such an investigation are presented, focusing primarily on the experimental advances of the in-beam γ-ray spectroscopy of proton-rich nuclei using RIBs and the state-of-the-art γ-ray spectrometer EXOGAM [1,2]. The EXOtic GAMma-ray spectrometer is the result of a European collaboration for the development of a high photopeak efficiency and energy-resolution γ-ray spectrometer. Its coupling with ancillary detectors provides a versatile and powerful tool for nuclear structure studies using RIBs, as demonstrated in this report.

2. In-beam studies using RIBs

The discovery of new exciting phenomena as well as the exploration of unknown territories on the Segré chart of nuclides have led to the investment of money, effort and time for the development of RIBs. However, the use of RIBs induces several experimental challenges, demanding the development of new spectrometers and the coupling of many complex detection systems. This is due to the low intensity of the RIBs, which requires highly efficient spectrometers for the observation of the nuclei of interest, produced with low cross-sections. Moreover, the high background due to the radioactive decay of the beam necessitates the use of a highly selective detection system, which can be achieved by the coupling of many detectors. Finally, the γ-ray spectrometers should present high granularity in order to compensate for the increased Doppler broadening effects when using inverse kinematics and close-packed geometries for greater photopeak efficiency.

3. Experimental details

The present experiment [3,4] was carried out at GANIL (Grand Accélérateur National d'Ions Lourds) [5], Caen, France where the SPIRAL (Système de Production d'Ions Radioactifs et d'Accélération en Ligne) facility [6,7] delivered a ^{76}Kr RIB at an energy of 4.34 MeV/u and with an average intensity of 5×10^5 particles per second. The RIB impinged on a target consisting of 1.1 mg/cm^2 of ^{58}Ni. This gave a rate in each HPGe crystal of 1–2 kHz. The irradiation time was around seven days. The detection system consisted of the EXOGAM γ-ray spectrometer coupled together with the DIAMANT charged-particle array [8–10] and the VAMOS recoil spectrometer [11,12].

The EXOGAM spectrometer is an array of high energy-resolution and photopeak efficiency segmented Clover germanium detectors and is used for nuclear structure studies at GANIL. An EXOGAM Clover detector consists of four coaxial n-type HPGe crystals housed in the same cryostat. Each crystal provides a high energy-resolution signal from the centre contact (<2.1 keV for 1.3 MeV γ rays) and is electrically segmented longitudinally on the outer boron-implanted p-type contact into four. The outer contact energy signals are poorer in energy resolution, but can yield a more precise determination of the position of a γ-ray interaction in the HPGe crystal. Each Clover detector is surrounded by an escape-suppression shield, comprising individual building blocks of bismuth germanate (BGO) and CsI(Na) scintillation detectors, which operates in four quadrants corresponding to the four HPGe crystals.

Depending on the experimental needs, the EXOGAM array, consisting of up to 16 Clovers with four detectors at 135°, eight detectors at 90° and four detectors at 45° with respect to the beam direction, can be arranged in different configurations [13]. In the present experiment EXOGAM consisted of 11 Clover detectors, with seven Clovers at 90° and four at 135°. The detectors were positioned at a distance of ~114 mm from the target position, using only the rear side shield (BGO) and the back catcher (CsI(Na)), in order to comprise a close-packed geometry. In this configuration (suppression-shield configuration A [13]), where the detectors can essentially touch at the front, the photopeak efficiency of the array is estimated to be ~12% at 1.33 MeV in addback mode,[2] i.e. summing the energy of neighbouring crystals that fire in an event [14].

The DIAMANT charged-particle array consisted of 48 CsI(Tl) scintillation crystals coupled to PIN diodes and provided channel selection by identifying light charged particles (protons, alphas) emitted from the compound system (^{134}Gd*). DIAMANT was mounted inside the target chamber with an almost 2π close-packed geometry covering forward angles with respect to the beam axis, from 8° to 90°.

The VAriable MOde Spectrometer (VAMOS) has been designed to identify reaction products induced by SPIRAL beams. It consists of two magnetic quadrupoles, a Wien filter, a magnetic dipole and the focal plane. In this experiment two secondary-electron detectors (SeD), an ionisation chamber and a plastic scintillator comprised the focal plane of the spectrometer. Due to a vacuum breakdown in the focal-plane detection system in the course of the experiment, the subset of the data that includes coincidences between γ rays and evaporation residues suffers from very poor statistics and thus the use of the spectrometer in the data analysis was precluded. Nonetheless, the quality of the γ-ray spectrum in coincidence with a fusion–evaporation residue in VAMOS [15] compares well with that measured with DIAMANT alone, demonstrating the capability of the spectrometer to handle kinematics for reactions such as ^{76}Kr on ^{58}Ni with low beam intensity.

The trigger condition used in the present experiment required an EXOGAM γ−γ prompt coincidence event, i.e. γ rays in two EXOGAM Clovers, or an EXOGAM–VAMOS coincidence event, i.e. a γ ray in EXOGAM and a recoil in VAMOS (any event that signals the two SeDs within a time window of 50–150 ns). DIAMANT was not included in the trigger and its signals, if any, were stored whenever any of the aforementioned triggers were satisfied. The data collected were merged using the GANIL CENTRUM[3] electronic module [16], providing time-stamped events before being written to disc for further off-line analysis.

4. Results

Approximately 10^9 events met the trigger conditions during the experiment and were recorded. Their corresponding γ-ray singles spectrum is shown in Fig. 1(a). This spectrum is dominated by the radioactivity of the ^{76}Kr beam, which decays via electron capture (EC) and β^+ emission to ^{76}Br and finally to the stable ^{76}Se nuclide [17]

$$^{76}\mathrm{Kr}(t_{1/2}=14.8\,\mathrm{h})\xrightarrow{\mathrm{EC},\beta^+}{}^{76}\mathrm{Br}(t_{1/2}=16.2\,\mathrm{h})\xrightarrow{\mathrm{EC},\beta^+}{}^{76}\mathrm{Se(stable)}.$$

The dominant peaks in the spectrum are the 511 keV annihilation peak and the 559 and 657 keV γ rays from the $2_1^+\to0^+$ and $2_2^+\to2_1^+$ transitions in ^{76}Se [17], respectively.

The present fusion–evaporation reaction produced recoiling residues[4] with a mean velocity[5] of $v=0.047c$, which resulted in a

[2] The photopeak efficiency of one crystal is measured to be ~0.16% at 1.33 MeV. The stated photopeak efficiency of the array in addback mode employs an addback factor that is derived from simulations.

[3] Clock Event Number Transmitter Receiver Universal Module.

[4] The strong exit channels in this reaction were the four-proton (4p) and three-proton (3p) channels leading to ^{130}Nd and ^{131}Pm, respectively. Transitions in the 5p (^{129}Pr), α3p (^{127}Pr), α2p (^{128}Nd) and α4p (^{126}Ce) exit channels were also observed.

[5] The mean recoiling-residue velocity is fairly approximated by the mean compound-nucleus velocity; small variations of ~5% for this value did not alter the quality of the γ-ray spectra.

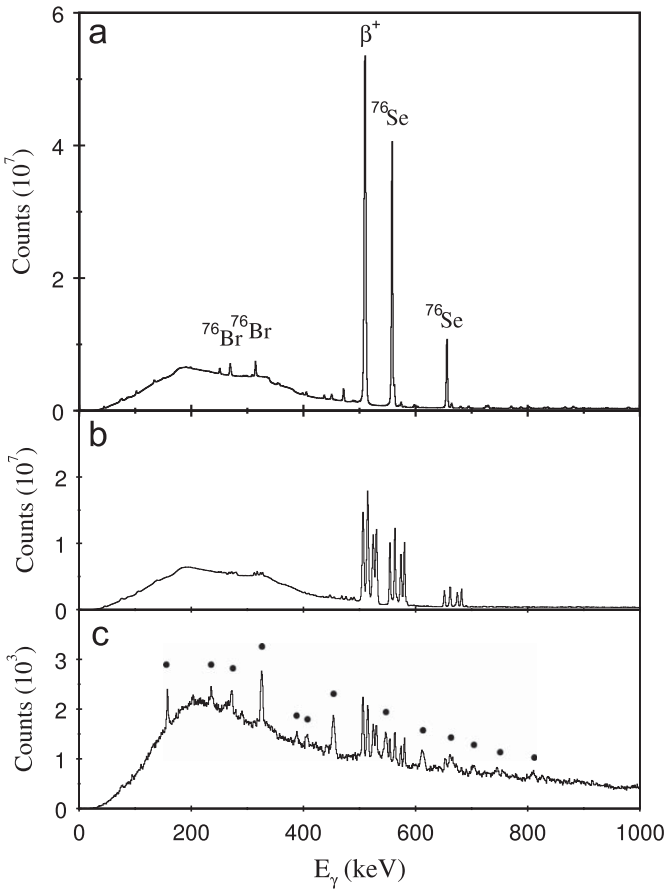

Fig. 1. (a) Total γ-ray spectrum (1 keV/channel). The labelled peaks correspond to the 511 keV annihilation peak, as well as to transitions depopulating excited states in ^{76}Br and ^{76}Se. (b) Total Doppler-corrected γ-ray spectrum (1 keV/channel). The background radioactivity peaks split into four, according to the four angles of the HPGe crystals with respect to which the γ rays are Doppler corrected. (c) Doppler-corrected γ-ray spectrum (1 keV/channel) in coincidence with at least one charged particle (proton, alpha) identified by DIAMANT. These events constitute only ~0.04% of the total number of events recorded during the experiment. γ rays from fusion products are now evident, labelled by solid circles.

Doppler shift and broadening of the detected γ rays. In order to correct for the Doppler effects, the angle of the HPGe crystal/segment with respect to the beam direction is required.[6] In Fig. 2 the sideview of an EXOGAM Clover detector is illustrated. Although its position with respect to the beam axis is at 90°, the crystals/segments have a slightly different angle. For the Doppler correction one needs to define this angle, which depends on the interaction depth of the γ rays in the HPGe crystals/segments.

The γ rays observed in this experiment are of energies lower than 800 keV and thus they interact with the material of the HPGe crystal within its first few centimetres. In Ref. [18] measurements on a prototype segmented Clover detector for the definition of this interaction depth are presented. Its optimum value for γ rays of energy up to ~600 keV has been deduced at ~30 mm. In the present experiment slight changes to the interaction depth around this value did not significantly alter the resolution of the γ-ray peaks. Thus the crystal/segment angles that were derived using

[6] The angle of the recoiling residue with respect to the beam direction should also be taken into account. However, in this experiment the inverse kinematics "set" the emitting angle of the proton-exit-channel residues at (almost) 0°. The residues in coincidence with alpha particles are emitted under a significantly larger angle thus inducing an additional broadening. In this report, however, the proton exit channels are of primary interest.

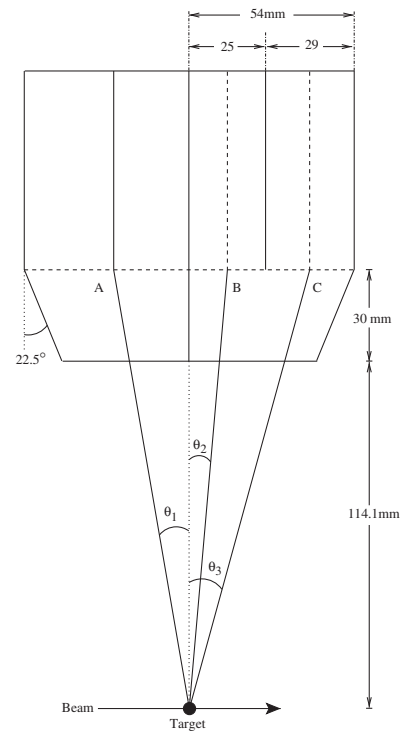

Fig. 2. Sideview of an EXOGAM Clover detector positioned at 90° with respect to the beam axis and centred at 114 mm from the target position. For an interaction depth of 30 mm at point A, the deviation from 90° of the crystal angle is $\theta_1 = 9.8°$. For an interaction depth of 30 mm at points B and C, the deviation from 90° of the segment angles is $\theta_2 = 5.0°$ and $\theta_3 = 15.3°$, respectively. Assuming that the recoiling nuclei travel along the beam direction, the crystal angle that should be used for the Doppler correction is $90° + \theta_1$, while the segment angle is $90° - \theta_{2,3}$.

Table 1
Defined angles for the EXOGAM crystals and segments considering an interaction depth of 30 mm.

Clover angle	Crystal angle	Segment angle
90°	$90° - \theta_1 = 80.2°$	$90° - \theta_3 = 74.7°$
		$90° - \theta_2 = 85.0°$
	$90° + \theta_1 = 99.8°$	$90° + \theta_2 = 95.0°$
		$90° + \theta_3 = 105.3°$
135°	$135° - \theta_1 = 125.2°$	$135° - \theta_3 = 119.7°$
		$135° - \theta_2 = 130.0°$
	$135° + \theta_1 = 144.8°$	$135° + \theta_2 = 140.0°$
		$135° + \theta_3 = 150.3°$

For this calculation the EXOGAM design illustrated in Fig. 2 is used, where the θ_1, θ_2 and θ_3 angles are introduced.

the EXOGAM design of Fig. 2 have been deduced assuming an interaction depth of 30 mm and are summarised in Table 1.

The γ-ray singles spectrum that is Doppler corrected according to the four HPGe crystals of each Clover is shown in Fig. 1(b). The background radioactivity peaks from the decay of the ^{76}Kr beam are split into four, since they are emitted from non-moving beam-like particles built up in material at and around the target position. In contrast, γ rays that are emitted from the moving fusion–evaporation products should now be aligned in energy.

4.1. DIAMANT gates

DIAMANT provided the charged-particle identification of the evaporation channels. For each element of DIAMANT three types

of signal are derived: the energy of the particle, its type (particle identification: PID) and a time-reference signal related to the time instance of the particle-γ coincidence [10]. The two-dimensional plot of the particle identification signal versus the energy (PID-Energy matrix), shown in Fig. 3(b), provides characteristic distinct patterns for each type of charged particle; the γ rays that interact with the CsI and the PIN diodes are also present. Before applying the two-dimensional gates on the detected protons and alphas, in order to suppress uncorrelated background events in DIAMANT, i.e. events that trigger DIAMANT but do not correspond to a prompt γ-particle coincidence, the timing information is extracted (PID-Time matrix in Fig. 3(a)). The true coincidences between fusion–evaporation charged particles and γ rays comprise a small fraction of the events signalled by DIAMANT, as can be seen by comparing Fig. 3(c) to (a). These events are obtained by narrowing the time gate of the γ-particle coincidence from 600 ns to approximately half this value and by selecting the appropriate PID and energy values that are expected for charged particles according to Fig. 3(b). The PID-Energy matrix that corresponds to these events is shown in Fig. 3(d); the final two-dimensional polygonal gates that are used to select the detected fusion–evaporation protons and alphas are also shown.

Considering the condition of emitted γ rays in coincidence with at least one charged particle (proton, alpha) in DIAMANT, i.e. pure fusion–evaporation events, the spectrum in Fig. 1(c) is obtained. These events constitute only a small fraction ~0.04% of the total number of events recorded during the experiment (Fig. 1(a) and (b)). It is evident that by gating on the charged particles detected by DIAMANT, most of the background radioactivity in the γ-ray spectra disappears.

The almost 2π solid angle coverage of DIAMANT reduces its absolute detection efficiency; however, this configuration was chosen to allow low-energy γ rays to be detected by EXOGAM, which covers backward angles, without significant absorption. Unfortunately, this results in the "contamination" of the charged-particle gated γ-ray spectra from exit channels with a greater number of particles emitted than that required by the gate, i.e. γ rays from the 3p and 4p exit channels will be visible in the 2p gated spectrum, as seen in Fig. 4. From the intensity ratios of such "contamination" peaks in the various proton-gated γ-ray spectra, the absolute proton-detection efficiency of DIAMANT has been measured to be $\varepsilon_p = 47(2)\%$ for this experiment [19].

4.2. Suppression shield

Events that deposit a fraction of their energy in the EXOGAM detector before scattering out can be suppressed in two ways. The first method, which is the most inefficient, uses the summed energy information from the suppression-shield elements of each

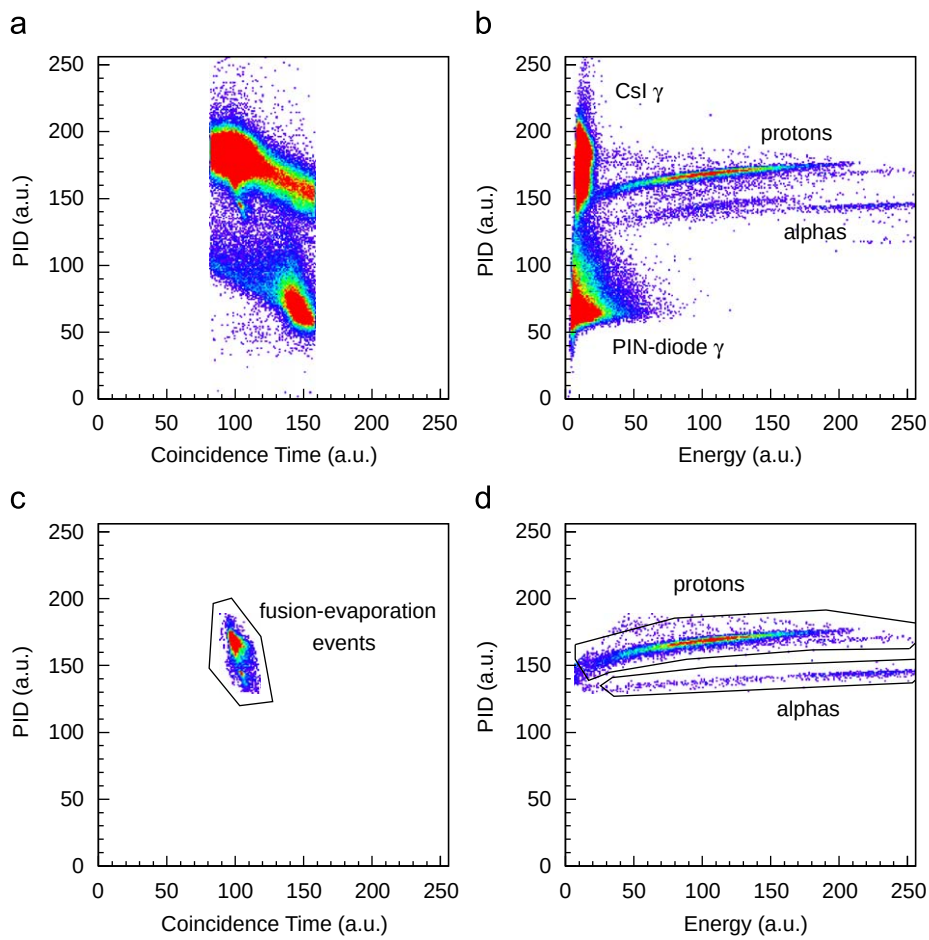

Fig. 3. (a) PID signal versus time of coincidence (PID-Time matrix). (b) PID signal versus energy (PID-Energy matrix). (c) PID signal versus time of coincidence for the prompt γ-particle coincidences. (d) PID-Energy matrix gated by the fusion-events cut of plot (c). All the uncorrelated events plus the γ rays detected by the CsI and the PIN diodes are suppressed. The two distinct patterns correspond to the protons and the alphas of the fusion exit channels. The two-dimensional polygonal gates that are used to select the detected protons and alphas are also shown.

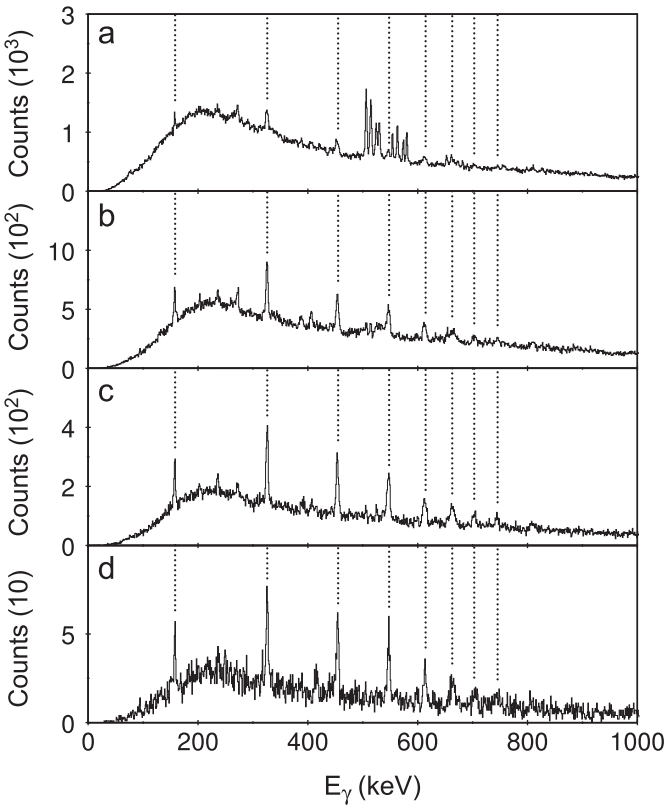

Fig. 4. Doppler-corrected γ-ray spectra (1 keV/channel) in coincidence with (a) one-proton emission, (b) two-proton emission, (c) three-proton emission and (d) four-proton emission. The dotted lines indicate transitions in the strongest four-proton exit channel ^{130}Nd.

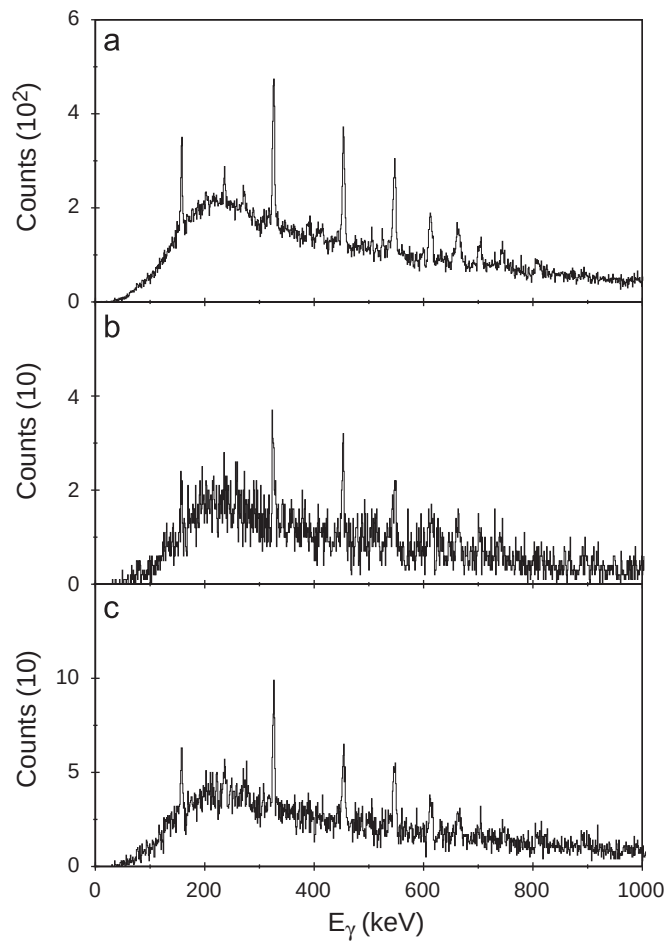

Fig. 5. γ-ray spectra (1 keV/channel) in coincidence with at least three-proton emission (a) without Compton suppression, (b) with Compton suppression, using the summed energy information from the shield, (c) with Compton suppression, using the "BGO pattern". The decrease of ~90% and ~75% in the statistics when suppressing the Compton-scattered events in spectra (b) and (c), respectively, compared with the non-suppressed γ-ray spectrum (a) is evident.

Clover and vetoes those γ events that are in coincidence with an event in the shield. The major disadvantage of this method is that the elements of the shield surrounding each crystal are not handled separately. In this way an event can be vetoed in crystal A for example when the shield surrounding crystal B has fired. This results in a dramatic decrease of ~90% in the statistics that can be seen in Fig. 5(b).

The second method of suppressing the Compton events is by using the "BGO pattern" of the so-called "quadrant suppression". This is a 12-bit word that corresponds to each element of the crystal and the shield, as shown in Table 2, and which is generated from the DAQ. By using this pattern, the Compton-suppression-shield information can be used separately for each crystal. The suppressed events involve a Ge bit for a specific crystal switched on (1) and the bits that correspond to the equivalent BGO and CsI switched off (0).

Analysing the data from the "BGO pattern", it turned out that the anti-Compton shield has fired for the majority of the events in this experiment and only ~25% of the events did not involve a shield element firing (Fig. 5(c)). However, it is not clear whether this is due to the high background radioactivity induced by the beam and built up at and around the target position or to the malfunctioning of the "BGO pattern". For this reason the suppression shield could not be reliably used in the data analysis.

Table 2
"BGO pattern" bit correspondence.

Signal	Bit	Signal	Bit	Signal	Bit
Ge(A)	1	BGO(A)	5	CsI(A)	9
Ge(B)	2	BGO(B)	6	CsI(B)	10
Ge(C)	3	BGO(C)	7	CsI(C)	11
Ge(D)	4	BGO(D)	8	CsI(D)	12

4.3. Segment-hit pattern

The idea of the "segment-hit pattern" is to use the angle information from the hits recorded in the segments and apply this to the energy given by the centre contact for each crystal, i.e. only the position information of the γ-ray interaction delivered by the segments is used.[7] In this experiment three scenarios are considered, constituting ~70% of the fusion–evaporation γ events according to Fig. 6, and which are illustrated in Fig. 7.

(1) Only one segment fires in the Clover. The energy from the centre contact, in which the fired segment corresponds, is corrected using the angle information from the fired segment. The corresponding γ-ray spectrum is shown in Fig. 8(a).

[7] A segment energy-calibration method has been developed for EXOGAM [20]. In this report, however, the energy information from the segments is not used.

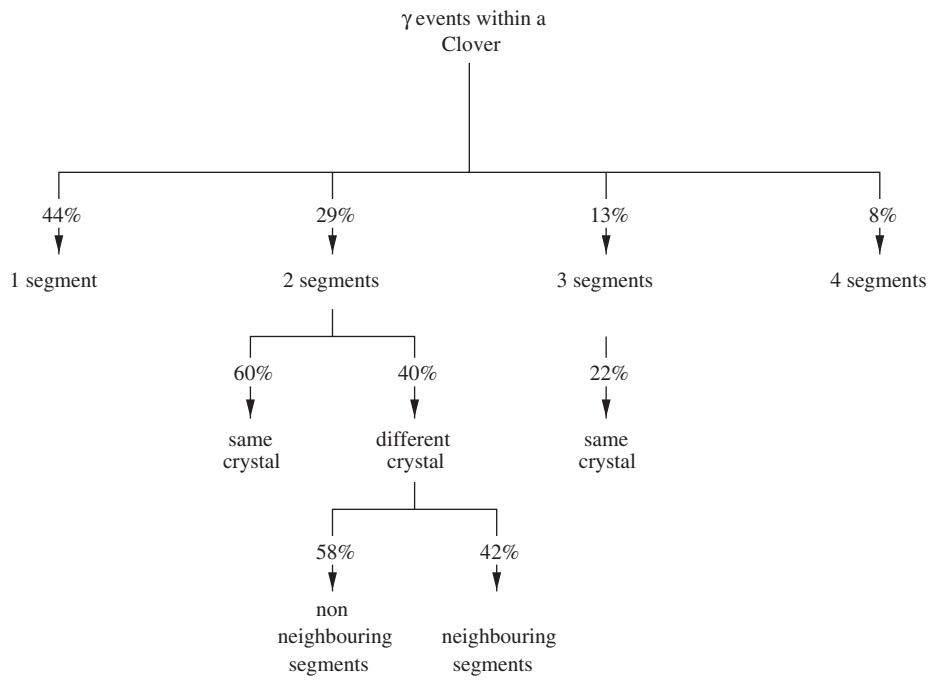

Fig. 6. Experimentally calculated percentages of the number of segments that fire for a γ event in an EXOGAM Clover when a highly selective DIAMANT gate is satisfied. These percentages are independent of the fusion–evaporation exit channel, i.e. at least 2p, 3p or 4p emission.

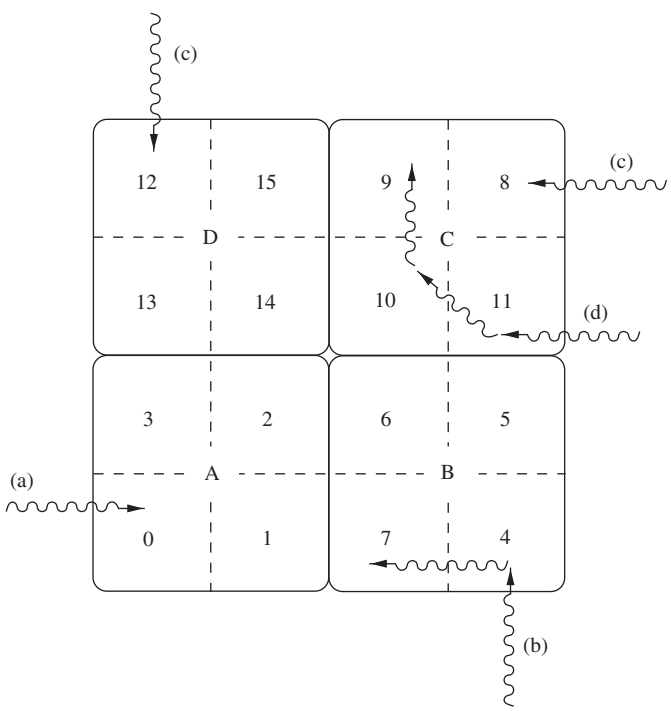

Fig. 7. Segment configuration within an EXOGAM Clover detector. The following cases are illustrated: (a) only one segment fires (segment 0), (b) two segments fire within the same HPGe crystal due to Compton scattering (segments 4 and 7), (c) two segments fire (segments 12 and 8), but they are not neighbouring, thus they are treated as different events and (d) three segments fire within the same HPGe crystal (segments 11, 10 and 9).

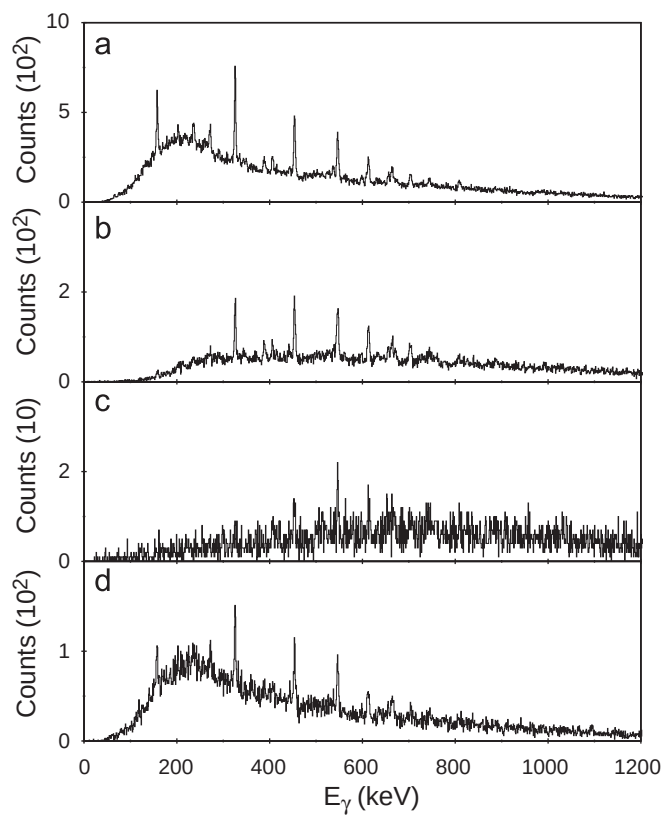

Fig. 8. γ-ray spectra (1 kev/channel) in coincidence with at least two-proton emission when (a) one segment fires, (b) two segments fire within the same crystal, (c) three segments fire within the same crystal and (d) two non-neighbouring segments fire in two different crystals. In (a) there is still background at low energy. This can be understood from events that either hit one segment and escape or events that hit more than one segment, but only one fires due to the high energy threshold that some segments present. Spectrum (d) presents even more background at low energy which is consistent with the fact that in the case of two firing non-neighbouring segments in different crystals, there is a percentage of the events that correspond to scatterings of the γ rays, but have not been identified as such.

(2) Two segments fire in the Clover. In this case one should further distinguish the following options:
   - Both segments come from the same crystal and the mean angle of the two segments, i.e. $(\theta_1 + \theta_2)/2$, is used to correct the energy centre-contact signal for the crystal. The

Table 3
The peak-to-background (P/B) ratio and the FWHM measured for three different energies and three exit channels with (ON) or without (OFF) the use of the "segment-hit pattern".

		158 keV		326 keV		704 keV	
Exit channel	Hit pattern	P/B	FWHM	P/B	FWHM	P/B	FWHM
At least 2p	OFF	0.24	3.6(2)	0.31	4.3(1)	0.18	7.6(8)
	ON	0.28	3.6(2)	0.39	3.5(1)	0.25	5.8(5)
At least 3p	OFF	0.32	3.1(2)	0.41	4.5(2)	0.26	7.2(9)
	ON	0.33	3.2(2)	0.50	3.4(1)	0.35	5.3(6)
At least 4p	OFF	0.47	2.6(5)	0.50	4.5(5)	0.48	8.3(16)
	ON	0.51	2.3(4)	0.58	3.8(4)	0.58	6.7(12)

γ-ray spectrum that matches these events is shown in Fig. 8(b).

- The two fired segments correspond to different crystals and in addition they are not neighbours.[8] In this case each hit can be assumed as independent from the other, and thus the energy centre-contact signal can be used for each crystal separately correcting with the corresponding fired-segment angle. This γ-ray spectrum is shown in Fig. 8(d).

(3) Three segments fire in the Clover. The simplest case of all three segments corresponding to the same crystal is taken into account and the angle of the crystal is used. Such a γ-ray spectrum is shown in Fig. 8(c).

In order to examine the improvement in the quality of γ-ray spectra when using the "segment-hit pattern", the peak-to-background (P/B) ratio and the resolution (FWHM) of the γ peaks have been calculated experimentally. The results are shown in Table 3. In this context the P/B ratio is defined as the ratio of the peak area to the total area under the peak (peak plus peak background). There is indeed a significant improvement in the γ-ray spectrum when using the "segment-hit pattern", as can be seen in Figs. 9–11, with the P/B ratio improvement reaching values as high as ~40%. This is due to the more accurate Doppler correction and the reduced opening angle of the detector because of the use of the segment information. Moreover, selecting events that do not exceed scattering into more than three segments reduces the background, as the energy range of the γ rays of interest does not support events that scatter within many segments. The deterioration of the FWHM for higher energies in the highly selective "at least 4p" DIAMANT gate, evident in Table 3, is due to low statistics.

4.4. Linear-polarisation measurements

The EXOGAM Clover detectors can act as Compton polarimeters [21,22] and γ-ray linear-polarisation measurements can be performed. The polarisation P of linear-polarised γ rays can be calculated as

$$P = \frac{A}{Q} \tag{1}$$

where A is the experimental asymmetry between parallel and perpendicular coincidence rates and Q, a function of γ-ray energy,

[8] For a first hit in crystal A and a second hit in B, C, or D, neighbours are characterised by the following combinations of segments 1–6, 1–7, 2–6, 2–7, 2–10, 2–13, 2–14, 3–13 and 3–14.

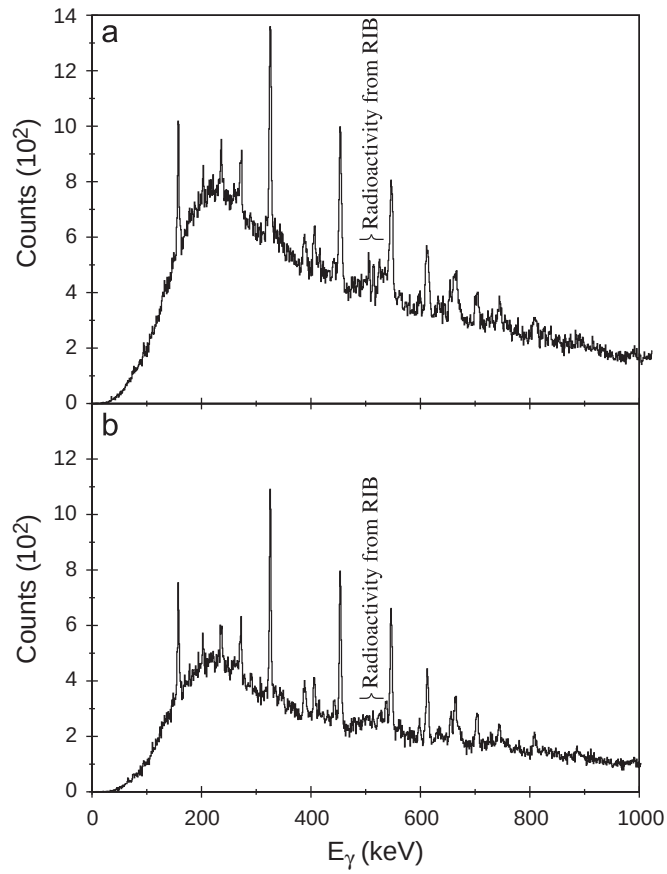

Fig. 9. γ-ray spectra (1 keV/channel) in coincidence with at least two-proton emission (a) without the "segment-hit pattern" and (b) using the information from the segments. There is an improvement in the peak-to-background ratio and in the resolution. Moreover, the background radioactivity peaks visible in (a) disappear in (b) due to the selectivity in the events that the "segment-hit pattern" demands.

is the polarisation sensitivity of the polarimeter. The experimental asymmetry is defined as

$$A = \frac{N_\perp - N_\parallel}{N_\perp + N_\parallel} \tag{2}$$

where $N_\perp$ and $N_\parallel$ are the number of events scattered perpendicular (⊥) and parallel (∥) to the reaction plane, as illustrated in Fig. 12.

The experimental asymmetry A of four well-known transitions has been measured and the results are shown in Table 4. For this measurement only the 90° Clovers (seven in total) have been used, since they are the most sensitive to the asymmetry [21], and a highly selective DIAMANT gate has been applied, i.e. at least two-proton emission. In order to perform such measurements, the energy of the γ ray under investigation should be large enough ($E_\gamma > 300$ keV) to observe its scattering between two segments, as seen experimentally in Fig. 13. Positive values of the asymmetry A are expected for stretched, purely electric transitions and negative values for stretched, purely magnetic transitions. The results of Table 4 act as a check on whether linear-polarisation measurements can be performed with such low beam intensity. Their agreement with the previously deduced multipolarities [23] prove that indeed such measurements have now become possible with the state-of-the-art EXOGAM array. It is, however, unfortunate that no magnetic transition could be measured.

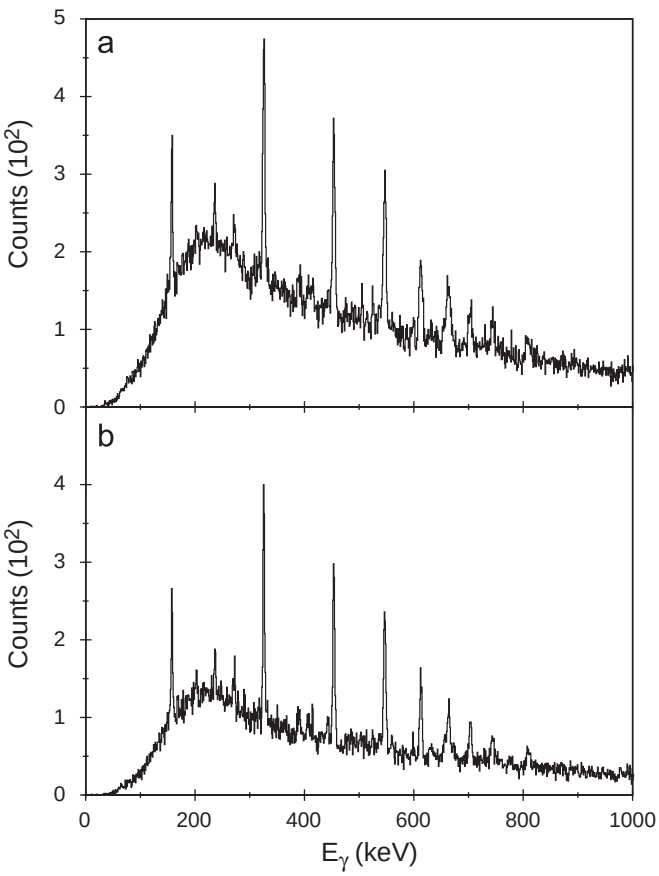

Fig. 10. γ-ray spectra (1 keV/channel) in coincidence with at least three-proton emission (a) without the "segment-hit pattern" and (b) using the information from the segments.

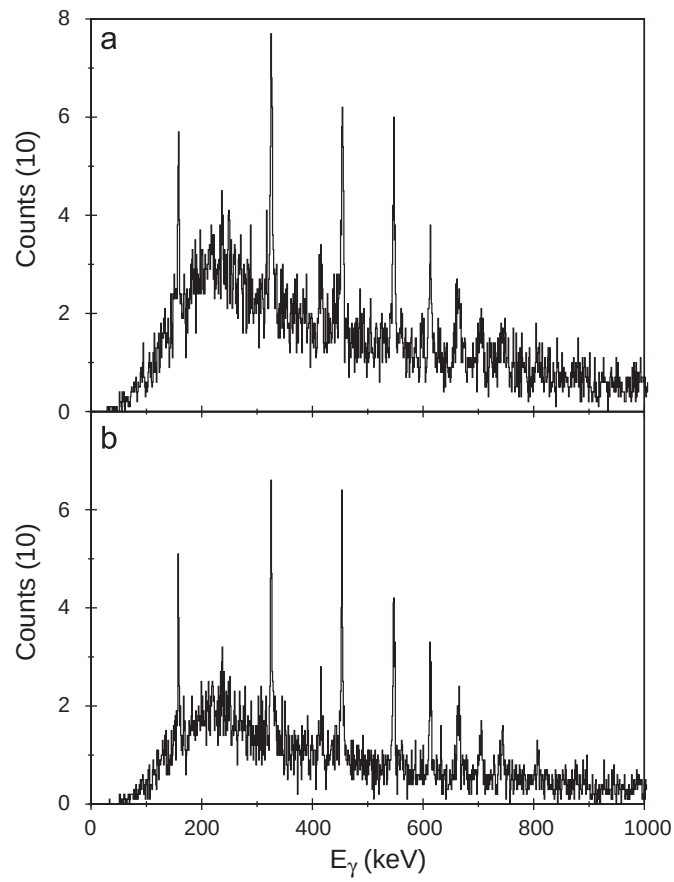

Fig. 11. γ-ray spectra (1 keV/channel) in coincidence with at least four-proton emission (a) without the "segment-hit pattern" and (b) using the information from the segments.

4.5. Cross-section sensitivity

Theoretical calculations using the ALICE code [24] for the cross-sections of the evaporation channels have allowed an estimation of the relative cross-section sensitivity of the present experimental setup according to the observed channels. This sensitivity is calculated by the theoretical cross-section of the evaporated channels multiplied by the γ-ray intensity of their highest transition observed in the γ-ray spectrum. The results are shown in Table 5, where a sensitivity of the order of 4 mb is estimated. However, this estimate is derived only from the observed even–even nuclides, ^{130}Nd, ^{128}Nd and ^{126}Ce, since the (relative) intensity of the highest γ-ray transition in the odd–even ^{131}Pm, ^{129}Pr and ^{127}Pr nuclides is not known. Nevertheless, their presence in Table 5 together with their ALICE calculated cross-section give a feeling of the cross-section sensitivity for the various populated/observed exit channels.

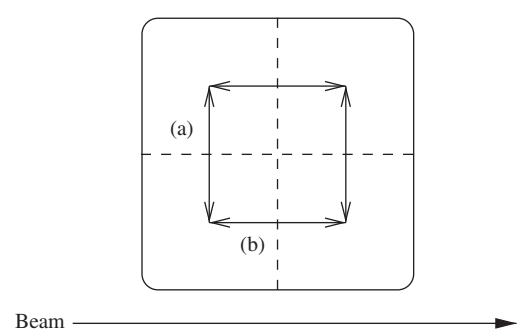

Fig. 12. Layout of the EXOGAM Clover crystal. Gamma events defining (a) perpendicular (⊥) and (b) parallel (||) to the reaction plane Compton-scattered events are illustrated. The beam direction is also shown.

5. Conclusion

The use of RIBs opens a whole new area in the spectroscopy of exotic nuclei which has not been studied yet. However, the induced background of the beam in combination with its low intensity imposes many experimental challenges. The EXOGAM γ-ray spectrometer in conjunction with ancillary detectors is a state-of-the-art tool for probing very exotic nuclei that are populated with RIBs. In the case of the ^{76}Kr + ^{58}Ni fusion–evaporation reaction, the strong 4p exit channel, ^{130}Nd, can be studied with the present experimental setup up to relatively high spin and even linear-polarisation measurements can be carried out. It is expected that spectroscopy down to a level of 5–10 mb can be performed, given that there is a valid trigger to select the fusion–evaporation events, i.e. a charged-particle detector such as DIAMANT to select the evaporated particles, and by improving the γ-ray spectrum with techniques such as the "segment-hit pattern". Further development of RIBs, namely higher intensity and a bigger variety of species, will

Table 4
Measured experimental asymmetry A for four transitions of the strong 4p exit channel ^{130}Nd.

E_γ (keV)	$I_i \rightarrow I_f$	Multipolarity	Asymmetry
326	$4^+ \rightarrow 2^+$	$E2$	0.03(9)
454	$6^+ \rightarrow 4^+$	$E2$	0.17(8)
548	$8^+ \rightarrow 6^+$	$E2$	0.03(8)
614	$10^+ \rightarrow 8^+$	$E2$	0.1(1)

The spin assignments and the multipolarities are taken from Ref. [23].

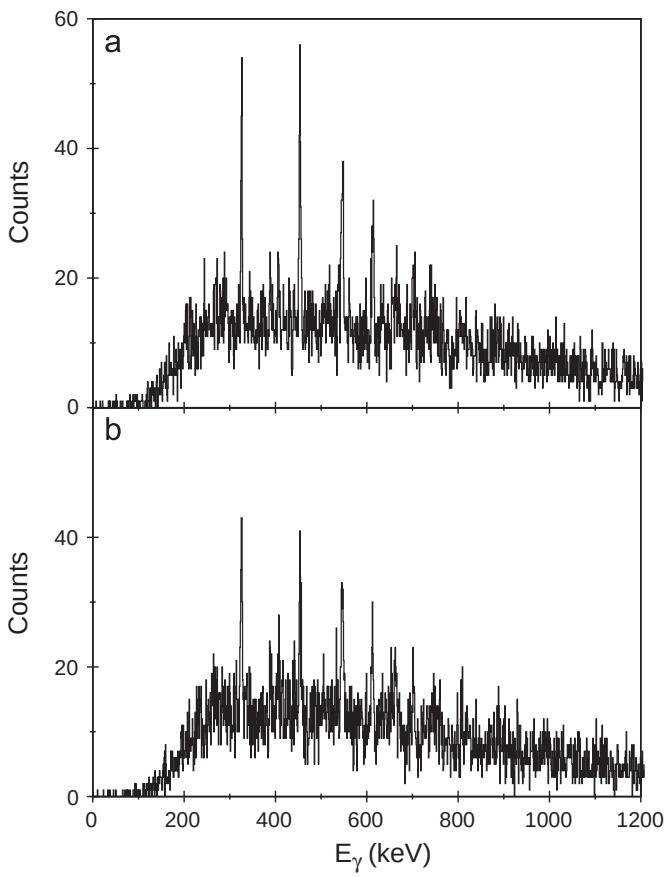

Fig. 13. γ-ray spectra (1 keV/channel) in coincidence with at least two-proton emission from the 90° Clovers, when two segments within the same crystal fire. γ rays scattered (a) perpendicular (⊥) and (b) parallel (∥) to the reaction plane are distinguished.

allow new mass regions to be reached, where the strong exit channels could lead to the discovery and study of completely unknown nuclei.

Table 5
Relative cross-section (σ) sensitivity using the ALICE theoretical cross-sections of the evaporated exit channels.

Observed channels		Highest spin	I_γ of highest observed transition (%)	ALICE σ (mb)	Relative σ sensitivity (mb)
3p	^{131}Pm	$27/2^-$	–	89	–
4p	^{130}Nd	22^+	2.7 [23]	164	4
5p	^{129}Pr	$27/2^-$	–	31	–
α2p	^{128}Nd	16^+	32 [25]	25	8
α3p	^{127}Pr	$35/2^-$	–	33	–
α4p	^{126}Ce	6^+	92 [26]	7	6

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